

Assignment 1.3

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Task 1

Fibonacci Sequence Generator (Without Functions)

Accept user input

n = int(input("Enter number of terms: "))

First two Fibonacci numbers

a = 0

b = 1

print("Fibonacci sequence:")

if n <= 0:

 print("Please enter a positive number.")

elif n == 1:

 print(a)

else:

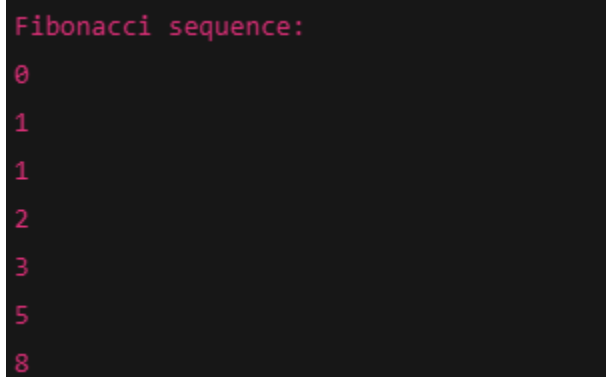
 print(a)

 print(b)

 count = 2

```
while count < n:  
    c = a + b  
    print(c)  
    a = b  
    b = c  
    count += 1
```

Output



```
Fibonacci sequence:  
0  
1  
1  
2  
3  
5  
8
```

Task 2

Optimized Fibonacci Sequence Generator

```
n = int(input("Enter number of terms: "))
```

```
a, b = 0, 1
```

```
print("Fibonacci sequence:")
```

```
for i in range(n):  
    print(a)
```

a, b = b, a + b

Output

```
Fibonacci sequence:
```

```
0
```

```
1
```

```
1
```

```
2
```

```
3
```

```
5
```

```
8
```

Task 3

Fibonacci Sequence Generator Using Functions

```
def fibonacci(n):
```

```
    """
```

```
    Generates and prints Fibonacci sequence up to n terms.
```

```
    """
```

```
    a, b = 0, 1
```

```
    for i in range(n):
```

```
        print(a)
```

```
        a, b = b, a + b
```

Main program

```
n = int(input("Enter number of terms: "))
```

```
if n <= 0:
    print("Please enter a positive number.")
else:
    print("Fibonacci sequence:")
    fibonacci(n)
```

Task 4

Procedural Fibonacci Program

```
n = int(input("Enter number of terms: "))
```

```
a, b = 0, 1
```

```
print("Fibonacci sequence:")
```

```
for i in range(n):
    print(a)
    a, b = b, a + b
```

Modular Fibonacci Program

```
def fibonacci(n):
    a, b = 0, 1
    for i in range(n):
```

```
print(a)
a, b = b, a + b
```

```
n = int(input("Enter number of terms: "))
```

```
print("Fibonacci sequence:")
fibonacci(n)
```

Task 5

1

Iterative Fibonacci Implementation

```
n = int(input("Enter number of terms: "))
```

```
a, b = 0, 1
```

```
print("Fibonacci sequence (Iterative):")
```

```
for i in range(n):
    print(a)
    a, b = b, a + b
```

2

Recursive Fibonacci Implementation

```
def fib(n):  
    if n <= 1:  
        return n  
    return fib(n - 1) + fib(n - 2)  
  
terms = int(input("Enter number of terms: "))  
  
print("Fibonacci sequence (Recursive):")  
  
for i in range(terms):  
    print(fib(i))
```

Output

```
0  
1  
1  
2  
3
```